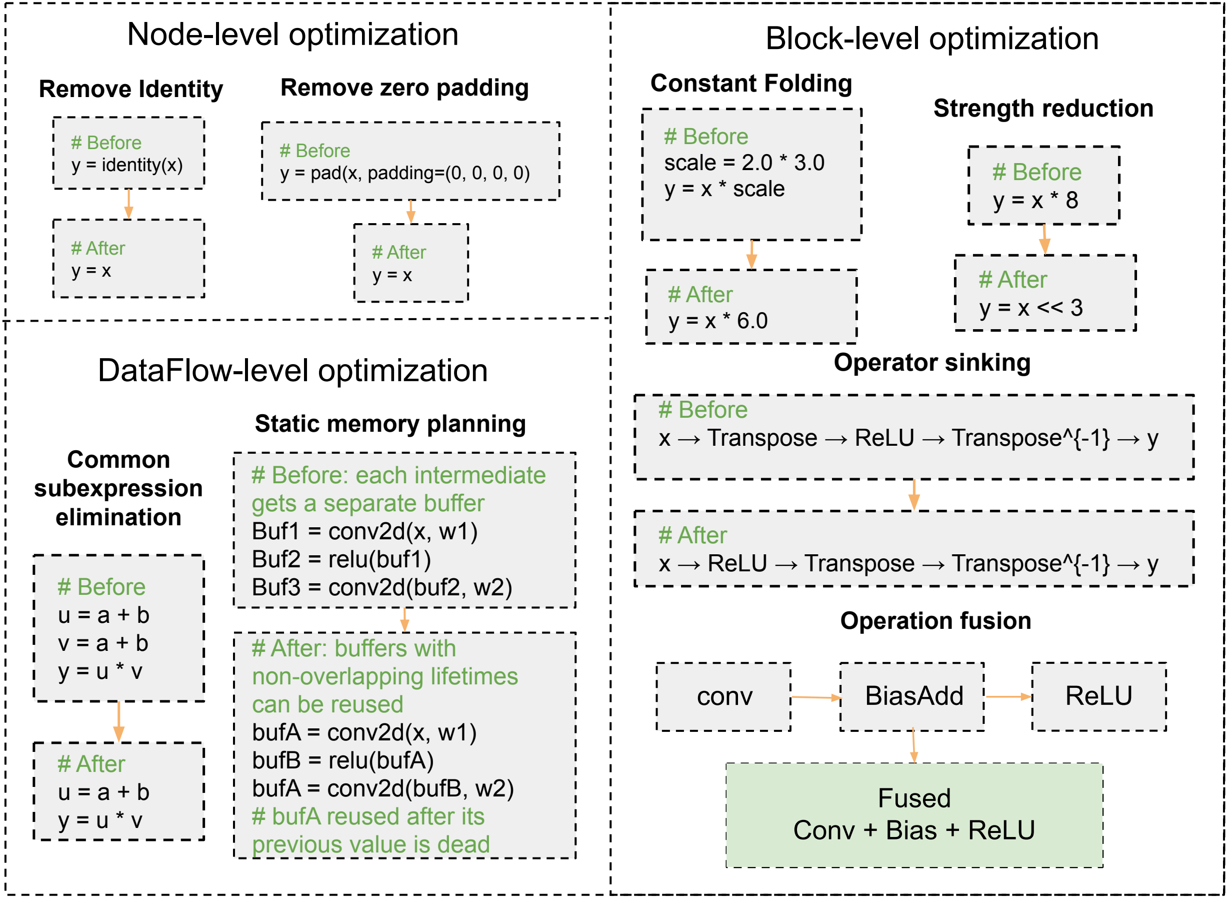

Node-level optimization
Remove Identity
# Before
y = identity(x)
# After
y = x
Remove zero padding
# Before
y = pad(x, padding=(0, 0, 0, 0)
# After
y = x
DataFlow-level optimization
Common subexpression elimination
# Before
u = a + b
v = a + b
y = u * v
# After
u = a + b
y = u * v
Static memory planning
# Before: each intermediate gets a separate buffer
Buf1 = conv2d(x, w1)
Buf2 = relu(buf1)
Buf3 = conv2d(buf2, w2)
# After: buffers with non-overlapping lifetimes can be reused
bufA = conv2d(x, w1)
bufB = relu(bufA)
bufA = conv2d(bufB, w2)
# bufA reused after its previous value is dead
Block-level optimization
Constant Folding
# Before
scale = 2.0 * 3.0
y = x * scale
# After
y = x * 6.0
Strength reduction
# Before
y = x * 8
# After
y = x << 3
Operator sinking
Before
x → Transpose → ReLU → Transpose^{-1} → y
After
x → ReLU → Transpose → Transpose^{-1} → y
Operation fusion
conv
BiasAdd
ReLU
Fused
Conv + Bias + ReLU